

A TRACE-73 CAPPELL–SHANESON HOMOTOPY 4-SPHERE DETECTED BY SKEIN LASAGNA MODULES

ANONYMOUS REPOSITORY AUDIT

ABSTRACT. We study the trace-73 Cappell–Shaneson construction associated with the matrix

$$A = \begin{pmatrix} 0 & 269 & 1240 \\ 0 & 41 & 189 \\ 1 & 0 & 32 \end{pmatrix}.$$

A public finite calculation produces the nonzero cubic Burau scalar -59072 in the grading convention corresponding to quantum degree 494. An exact Artin–Magnus certificate and the pure-braid Andreadakis theorem establish a third-order property of the public braid. We construct a public Aitchison–Rubinstein product-annulus presentation with an actual embedded detector collar and an action-compatible Hattori coefficient equivalence. The induced quantum trace has a divided cubic functional that annihilates all two-handle relations and is nonzero in quantum degree 494. A relative complete-sphere argument proves the three-handle closure, while published four-handle and standard-sphere results show that the resulting homotopy sphere is not diffeomorphic to S^4 . The Lean development checks the finite and quotient algebra; it does not formalize four-dimensional differential topology.

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1. INTRODUCTION

The smooth four-dimensional Poincare conjecture asks whether every smooth closed manifold homotopy equivalent to S^4 is diffeomorphic to S^4 . Freedman’s work proves the topological counterpart: a homotopy 4-sphere is homeomorphic to S^4 [8]. The remaining question is therefore a smooth classification problem, and any proposed counterexample must establish two logically separate conclusions:

- (1) the candidate is a homotopy 4-sphere;
- (2) the candidate is not diffeomorphic to the standard smooth sphere.

Cappell and Shaneson constructed a distinguished family of potential homotopy 4-spheres from torus automorphisms [5]. Their handle structures were studied by Aitchison and Rubinstein [1]. Many subfamilies were subsequently shown to be standard, through work including Akbulut [2], Gompf [9], Kim–Yamada [12], and Iwaki [11]. This history makes geometric identification especially important: agreement of a matrix, a relator, or a trace parameter is not a substitute for identifying a complete framed handle presentation.

Skein lasagna modules extend Khovanov–Rozansky link homology to smooth four-manifolds [15]. Manolescu and Neithalath gave a two-handlebody description [13], and Manolescu, Walker, and Wedrich gave formulas compatible with general handle decompositions [14]. These results provide an attractive route to conclusion (ii): a nonzero class in a nonzero bidegree

cannot be carried by a diffeomorphism to the lasagna module of S^4 , which is concentrated in bidegree zero.

We construct the public finite class for the standard-form Cappell–Shaneson parameter $(41, 189, 73)$. Compact product-ribbon, point-push, owner-lift and Nielsen ledgers replace parts of the unavailable historical data. The public AR product witness supplies the embedded framed Kirby presentation and detector collar. The product-annulus pairing supplies the candidate-specific quantum-trace map, and a standard sphere system disjoint from the detector gives the three-handle closure.

Contributions. The main contributions are as follows.

- (1) We give a public parameterized Aitchison–Rubinstein product-annulus witness, two geometric cancellations, and an embedded six-sweep detector collar, thereby replacing the unavailable historical PD.
- (2) We construct the action-compatible MWW Hattori comparison, identify the local checked R-matrix with the public Burau blocks, and prove the complete divided beta/psi cocone.
- (3) We use the relative uniqueness of complete sphere systems and MWW’s intrinsic local module action to derive the three-handle closure from P0 and the symmetric-monoidal form of C, without TH-sized sphere movies.
- (4) We provide executable ledgers, mutation tests and a kernel-checked Lean algebraic core.

Historical files referenced by earlier versions of the project are not publicly available. Their hashes are retained as provenance. The public AR, coefficient-comparison, and relative-sphere witnesses replace their load-bearing roles.

2. THE TRACE-73 CAPPELL–SHANESON CANDIDATE

Let

$$(1) \quad A = \begin{pmatrix} 0 & 269 & 1240 \\ 0 & 41 & 189 \\ 1 & 0 & 32 \end{pmatrix}.$$

This is Iwaki’s standard form $X_{41,189,73}$; the triple occurs among the trace-73 representatives in his table [11, Table 2]. Direct integer arithmetic gives

$$(2) \quad \det A = 1, \quad \det(A - I) = 1.$$

For $A \in \mathrm{SL}(3, \mathbb{Z})$, let $f_A : T^3 \rightarrow T^3$ be the induced diffeomorphism, made equal to the identity near a chosen point. Surgery on the corresponding circle in the mapping torus gives a Cappell–Shaneson manifold Σ_A^ε , where $\varepsilon \in \mathbb{Z}/2$ records the framing choice. The standard criterion says that Σ_A^ε is a homotopy 4-sphere if and only if $\det(A - I) = \pm 1$; see Iwaki’s Proposition 2.1 and the original sources cited there [11, 5].

Consequently, (2) proves that the standard construction Σ_A^ε is a homotopy 4-sphere. It does not prove that a separately generated large Kirby diagram

is a diagram of that construction. That identification is the first and most load-bearing hypothesis below.

Iwaki proves standardness for many infinite families. The appearance of $(41, 189, 73)$ in a representative table is not a standardness or exoticness theorem for this individual sphere. Likewise, the trace bound in Kim–Yamada’s work stops below 73 [12]. These observations explain the choice of candidate but do not decide its diffeomorphism type.

3. PRECISE STATEMENTS

We first state the abstract theorem in the form actually proved by the Lean development. This prevents geometric intuition from silently strengthening the result.

Fix a collection of manifold labels, a graded family of rational vector spaces $G(M, q)$, labels X and S^4 , and predicates $\text{HS}(M)$ and $\text{Diff}(M, N)$.

Theorem 3.1 (Abstract nonstandardness theorem). *Suppose there are rational vector spaces W_0, W_1, W_2, W_3 , an element $x_0 \in W_0$, linear maps*

$$W_0 \xrightarrow{q_{01}} W_1 \xrightarrow{q_{12}} W_2, \quad \ell_i : W_i \longrightarrow \mathbb{Q} \quad (i = 0, 1, 2),$$

and linear isomorphisms

$$T : W_2 \xrightarrow{\cong} W_3, \quad F : W_3 \xrightarrow{\cong} G(X, 494),$$

such that

$$(3) \quad \ell_0(x_0) = -59072,$$

$$(4) \quad \ell_1 q_{01} = \ell_0,$$

$$(5) \quad \ell_2 q_{12} = \ell_1.$$

Assume also that every element of $G(S^4, 494)$ is zero and that every diffeomorphism $X \rightarrow S^4$ induces a linear isomorphism $G(X, 494) \cong G(S^4, 494)$. Then X is not diffeomorphic to S^4 .

Proof. Set $v = F(T(q_{12}(q_{01}(x_0))))$. If $v = 0$, injectivity of F and T implies $q_{12}(q_{01}(x_0)) = 0$. Applying ℓ_2 and using (4)–(5) gives $-59072 = 0$, a contradiction. Thus $v \neq 0$. If X were diffeomorphic to S^4 , the induced isomorphism would send v to an element of $G(S^4, 494)$, hence to zero, contradicting injectivity. $\square$

The Lean theorem corresponding to Theorem 3.1 accepts the input structures `ExternalGeometry` and `CSExternalGeometry`; no inhabitants of these structures are constructed in the repository. We name the candidate-specific assumptions needed to instantiate them.

Theorem 3.2 (Public AR presentation, P0). *The public AR product witness is a labelled and framed handle presentation of Σ_A^0 . Two product cancellations give the registered genus-two attaching link. A specified boundary 3-ball B contains an actual six-sweep detector collar inducing the public 44- and*

88-strand braids on all oriented cable states. The presentation has three 3-handles followed by one final 4-handle.

Theorem 3.3 (Coefficient comparison, C). *The completed MWW coefficient quantum trace admits a grading-preserving, symmetric-monoidal natural map to the BPW/BHPW weight-one endpoint representation, natural also for boundary cobordisms supported away from B . Under this map the point-push action is the public Burau representation, and the divided cubic row descends through every beta and psi relation. Its selected value is -59072 in the public normalization; all strict pivotal sign variants are nonzero. The class lies in quantum degree 494.*

Theorem 3.4 (Relative three-handle consequence, S). *Under Theorems 3.2 and 3.3, the three 3-handles may be attached along a complete sphere system disjoint from B . For each sphere the detector kills the undotted MWW relation and identifies the once-dotted map with the source term. Hence the divided detector descends through the complete three-handle coequalizer.*

Theorem 3.5 (Final identifications, P3). *MWW’s three-/four-handle formulas identify the final quotient with $\mathcal{S}_{0,(0,494),0}^2(\Sigma_A^0; \mathbb{Q})$. The corresponding summand for S^4 is zero, and diffeomorphisms induce grading-preserving equivalences. Moreover, $\det(A - I) = 1$ makes Σ_A^0 a homotopy sphere.*

Theorem 3.6 (Main theorem). *The trace-73 Cappell–Shaneson manifold satisfies*

$$\Sigma_A^0 \simeq S^4 \quad \text{and} \quad \Sigma_A^0 \not\approx_{\text{diff}} S^4.$$

Proof of Theorem 3.6. Theorems 3.2 and 3.5, and the Cappell–Shaneson criterion give the homotopy-sphere assertion. Theorem 3.3 supplies the coefficient comparison, and Theorem 3.4 supplies the three-handle descent. They instantiate Theorem 3.1, whose selected class is nonzero in quantum degree 494. MWW’s four-handle isomorphism preserves this degree, while the same degree of the standard S^4 module vanishes. Diffeomorphism invariance and Theorem 3.1 rule out a diffeomorphism to S^4 . $\square$

4. THE FINITE COMPUTATIONAL LAYER

The public finite layer has three independent parts: lattice arithmetic, a chosen-sphere coordinate determinant, and a truncated Burau calculation.

4.1. Integral lattice calculations. Besides (2), the repository records the proposed sphere coordinate columns

$$(6) \quad C = \begin{pmatrix} -1311 & -189 & 41 \\ 8608 & 1241 & -269 \\ -1 & 0 & 1 \end{pmatrix}, \quad \det C = 1.$$

Thus the three recorded coordinate vectors form a basis of $\mathbb{Z}^3$. This is only the last arithmetic step in a geometric-basis argument. It does not construct embedded spheres, prove disjointness, or identify $\mathbb{Z}^3$ with $H_2^{\text{sph}}(\partial W_2)$.

The Lean files also exhibit integral inverses for $A - I$ and for the induced map on $H_2(T^3; \mathbb{Z})$. These computations eliminate the candidate-specific lattice algebra from the homotopy-sphere branch. Van Kampen, Wang–Mayer–Vietoris, Poincare duality, and Hurewicz–Whitehead arguments remain supplied through a topology structure rather than formalized as four-manifold topology.

4.2. The public Burau computation. Let $\varepsilon = t - 1$. The script `scripts/recompute_t73_delta3.py` reads a result-free JSON input and performs the following finite operations:

- (1) parse 252 primitive crossing rows and reconstruct an 11,340-letter Artin word on 44 strands;
- (2) cable it to a 45,360-letter word on 88 strands;
- (3) apply the unreduced Burau representation over $\mathbb{Z}[\varepsilon]/(\varepsilon^7)$;
- (4) substitute $t = q^{-2}$ with $q = 1 + h$, equivalently $\varepsilon = (1 + h)^{-2} - 1$;
- (5) extract the coefficient of h^3 after the specified source vector and covector are applied.

Denote the resulting scalar by δ_3^{Bur} . The independently reproduced public result is

$$(7) \quad \delta_3^{\text{Bur}} = -59072 = (-2)^3 \cdot 7384.$$

The public input does not contain either decimal literal 7384 or -59072 ; the former is, however, stored as the constant `cubicBase` in the Lean finite layer. Lean checks only the arithmetic $(-2)^3 \cdot 7384 = -59072$ and never evaluates the 45,360-letter word.

Proposition 4.1 (Exact scope of the computation). *Equation (7) is a statement about the specified truncated Burau model. The script and JSON alone do not define the MWW invariant. Under the coefficient-trace comparison of Theorem 3.3, however,*

$$\delta_3^{\text{MWW}} = \delta_3^{\text{Bur}} = -59072,$$

up to the harmless global convention unit normalized in that theorem.

Proof. The script defines vector and covector operations in an 88-dimensional truncated polynomial representation. The displayed MWW equality is exactly the additional content of Hypothesis 3.3; it does not follow from the finite script or from the local Burau-block calculation. $\square$

Thus the finite recomputation and the categorical comparison have distinct roles: the former supplies the integer, while the latter makes it a four-manifold invariant.

5. LASAGNA MODULES AND THE INTENDED OBSTRUCTION

We recall only the features needed for the comparison. For a smooth compact oriented four-manifold W and a framed link $L \subset \partial W$, the skein

lasagna module $\mathcal{S}_0^N(W; L)$ is generated by decorated surfaces in W , modulo local relations coming from KhR_N cobordism maps [15, 14].

For a handle decomposition

$$W_1 \subset W_2 \subset W_3 \subset W_4,$$

where W_1 is a one-handlebody, W_2 adds two-handles along a framed link, W_3 adds three-handles along embedded spheres, and W_4 adds four-handles, the published results have the following roles.

- MWW Theorem 3.2 identifies the two-handle stage with a cabled skein lasagna quotient, whose relations include beta and psi operations.
- MWW Theorems 3.7 and 3.10 describe each three-handle as a co-equalizer of the two hemisphere maps, and give the complete handle formula.
- MWW Proposition 3.4 says that a four-handle attachment induces an isomorphism for the empty boundary link.
- MWW Corollary 3.5 gives $\mathcal{S}_0^N(S^4) \cong \mathbb{Z}$, concentrated in bidegree $(0, 0)$.

At $N = 2$, a nonzero class in bidegree $(0, 494)$ would therefore obstruct a diffeomorphism to S^4 . The intended proof constructs a functional at the one-handle stage, shows that it annihilates the two-handle and three-handle relations, and transports the resulting nonzero class through the four-handle isomorphism.

The formal algebra correctly captures the last sentence. The next sections construct and verify its candidate-specific geometric and functorial inputs.

6. COMPACT PRODUCT-RIBBON PRESENTATION

We prove Theorem 3.2. The construction below replaces, rather than reconstructs, the unavailable historical planar diagram.

6.1. The actual AR link. Use the coordinate-spine genus-three Heegaard splitting of T^3 from Aitchison–Rubinstein [1, pp. 5–12]; the complete volume is available in a [public Berkeley scan](#). Let C_i be the three coordinate spine circles with common base vertex Q , and let A_i be pairwise-disjoint narrow disk neighborhoods away from Q . Write t for the mapping-torus one-handle. If ψ_A is the handlebody-preserving representative supplied by their Lemma 3.1, the core of the i th mapping-torus two-handle is the embedded circle

$$(8) \quad C_i^- \cup \lambda_i \cup \psi_A(C_i)^+ \cup \mu_i,$$

where λ_i, μ_i are the two arcs through t . The framing annulus is constructed simultaneously as

$$(9) \quad A_i^- \cup (\lambda_i \times I) \cup \psi_A(A_i)^+ \cup (\mu_i \times I).$$

The disks, cones and product levels in the AR construction are pairwise disjoint, so (8)–(9) define the whole embedded framed link, not merely its words.

The remaining three fiber two-handles are the standard dual two-cells of the coordinate-spine splitting. After choosing orientations, denote their boundary curves by

$$r_{xy} = [x, y], \quad r_{yz} = [y, z], \quad r_{zx} = [z, x].$$

Their embeddings are defined by the dual cells and their framings are the fiber product framings. The untwisted section-surgery handle h_{CS} goes geometrically once over t . Thus the starting handle counts are $(1, 4, 7, 3, 1)$ in indices $0, \dots, 4$: AR p. 8 removes one 3- and the 4-handle from the mapping torus and glues $S^2 \times D^2 = H^2 \cup H^4$ in their place.

For an independently checkable standard-form bridge, set

$$C_{AR} = \begin{pmatrix} 0 & 0 & 1 \\ 713 & 83 & 0 \\ -902 & -105 & -10 \end{pmatrix}, \quad R = \begin{pmatrix} 71 & -466 & 1 \\ -610 & 4004 & -7 \\ 1 & -7 & -2 \end{pmatrix}.$$

Direct multiplication gives

$$\det R = 1, \quad C_{AR}R = RA,$$

and both matrices have determinant one and determinant-one difference from the identity. Hence the fiber diffeomorphism induced by R transports the complete AR construction to the displayed matrix A . Since $R \in GL^+(3, \mathbb{R})$ and this group is connected, its derivative preserves the homotopy class of the canonical normal framing of the section circle; in particular, the untwisted choice $\varepsilon = 0$ is preserved.

6.2. Return to the linear torus map. Let ϕ_A be the linear torus map, straightened to the identity on the section ball, and choose an isotopy ρ_u , $u \in [-1, 1]$, from ϕ_A to ψ_A , stationary near its endpoints and relative to that ball. Put $g_u = \rho_u \phi_A^{-1}$. With mapping-torus convention $(x, -1) \sim (\phi(x), 1)$, the formula

$$(10) \quad F([x, u]_{\phi_A}) = [g_u(x), u]_{\psi_A}$$

defines a diffeomorphism. Indeed, $g_{-1} = \text{id}$, $g_1 = \psi_A \phi_A^{-1}$, and therefore $\psi_A g_{-1} = g_1 \phi_A$, which is precisely well-definedness at the seam. Endpoint stationarity makes (10) smooth, and the family g_u^{-1} gives its inverse. It fixes the section ball. Pulling the entire AR handle decomposition back by F replaces the top cores by the linear images $\phi_A(C_i)$ and transports every component and framing annulus simultaneously.

6.3. Words and normal fields. Lift C_i and $\phi_A(C_i)$ to the universal cover. The three top lifts begin at the same point $A(Q)$. Translate the whole top fiber by $-A(Q)$ and extend this translation over its collar. It is isotopic to the identity and transports the entire link and its normal fields. Since each attaching core is unbased, changing the first face crossing only cyclically permutes its word. We may therefore take the top centerline to be the straight segment from 0 to Ae_i ; it crosses the j th integer faces at the rational times $k/(Ae_i)_j$. A fixed arbitrarily small product perturbation orders simultaneous

events. Reading the two base arcs and the oppositely oriented bottom core gives exactly

$$(11) \quad m_i = t\phi_A(x_i)t^{-1}x_i^{-1}.$$

This is the full event rule used by the public generator.

The framing is equally explicit. AR push the coordinate axes in the universal cover in direction $(1, 1, 1)$; the resulting strips have parallel boundary components, and linearity of A preserves this property on their images [1, pp. 16–17]. Together with the λ_i/μ_i rectangles, these strips are the normal fields in (9). Thus no blackboard integer is substituted for a framing.

6.4. The two cancellations. Aitchison–Rubinstein identify (t, h_{CS}) as a complementary geometric one/two-handle pair [1, p. 7]. Slide every other two-handle passage over h_{CS} along its product rectangle and cancel; this removes the t, t^{-1} passages with zero relative twist.

Now $Ae_1 = e_3$, so the actual curve m_1 becomes zx^{-1} and has one geometric passage over the x one-handle. Thus (x, m_1) is another complementary pair. Sliding every other x passage over m_1 along parallel product bands replaces x by z and cancels the pair. All labels and normal fields move with the bands. The resulting counts are $(1, 2, 5, 3, 1)$, and the exact word projection is

$$\begin{aligned} |m_2| &= 311, & [m_2]_{(y,z)} &= (40, 269), \\ |m_3| &= 1460, & [m_3]_{(y,z)} &= (189, 1271). \end{aligned}$$

The transported r_{zx} has empty free word, but no split-unknot conclusion is needed or inferred from that fact.

6.5. The actual detector ball. Cut the remaining genus-two handlebody along its y, z disk system. The selected m_2/r_{xy} state has 42 y passages from m_2 and two from r_{xy} . At distinct product-normal levels, a regular neighborhood of these 44 passages is a standard three-ball B containing a punctured-disk collar. The six affine sweeps specify a pure 44-strand braid in this collar. The mapping-class interpretation of the braid group realizes it by an isotopy of the punctured disk, and isotopy extension gives an ambient isotopy of B . Choose the extension to be the identity to first order near every returned puncture at its terminal time. Hence each product normal returns, not merely the total writhe. The motion acts on every oriented normal copy simultaneously. The public verifier regenerates all 252 factors and obtains

$$|B_{44}| = 11340,$$

$$\text{SHA256}(B_{44}) = \begin{array}{c} 7C2D2F792C2672221A76CAF08A71F560A \\ F0CB7654B4D537C00FAD00B16EFA187 \end{array}.$$

$$\text{perm}(B_{44}) = 1, \quad \text{wr}(B_{44}) = 0.$$

The exact smooth construction, source digest, matrix bridge, handle counts, cancellation data and collar labels are regenerated in `audit/t73_`

`ar_product_witness.json`. Its digest is

AA9328EA3C7A53B937C97A02D611850D
0E8EEAF87C68756C04B013A486F6F112.

The source-PDF, regeneration and mutation checks are described in the repository README. This proves Theorem 3.2 without identifying the retired historical PD.

7. THE COEFFICIENT-TRACE COMPARISON

We prove Theorem 3.3. The strength needed here is precise: a homogeneous quantum-trace map is constructed at the one-handle stage, while its leading divided cubic, rather than the entire formal series, descends through the two-handle quotient.

7.1. The product Hattori equivalence. The compact words give

$$p_y = 42 + 2 = 44, \quad p_z = 269 + 2 = 271.$$

More is true than this count. In the cyclic m_2 word, every one of the 42 y/Y events is followed immediately by a distinct z/Z event; the same holds for the two y/Y events of r_{xy} . The witness lists all 44 pairs. The corresponding one-edge intervals on the actual P0 product annuli are pairwise-disjoint subrectangles. Their removal leaves exactly 227 unpaired z passages, closing to separate product circles between the two parallel annulus boundaries.

Let $\mathcal{C}$ be the pregraded rational tangle category with objects $T : P_{86} \rightarrow P_{88}$, and let F be the framed west-to-east tangle formed by the 44 rectangles. MWW's one-handle theorem identifies the cut coefficient as

$$M_R(T, T') = \text{KhR}_2(R \cup T' \cup \overline{T})$$

with its displayed shift and gluing actions [14, Theorem 4.7]. The fixed product isotopy, pivotal tangle duality, and Künneth over $\mathbb{Q}$ give graded isomorphisms

(12)

$$H_{T, T'} : M_R(T, T') \xrightarrow{\cong} \text{Hom}_{\mathcal{C}}(FT, FT')\{-44\} \otimes A^{\otimes 227}, \quad A = \mathbb{Q}[X]/(X^2).$$

They form a coefficient-bimodule isomorphism. Indeed, gluing $f : S \rightarrow T$ before the source link and then applying the fixed product isotopy is isotopic rel boundary to first applying the isotopy and precomposing by Ff . The right action is the same argument with postcomposition. Thus both action squares are associativity of gluing for one simultaneous surface.

Apply the 227 disjoint counits to (12). This is a homogeneous coefficient morphism. The remaining coefficient $(T, T') \mapsto \text{Hom}(FT, FT')$ is representable by the invertible tangle F and hence is equivalent to the regular coefficient. The ordinary quotient statement, including both cyclic-relation spans, is kernel checked in `Smooth4PC/RepresentableCoefficient.lean`.

7.2. Quantum trace and the endpoint map. Set $R_q = \mathbb{Q}[q, q^{-1}]$. The pregraded cyclic relation is

$$L_f(m) = q^{|f|} R_f(m).$$

Because (12) and the counits are homogeneous coefficient morphisms, they carry each such relation to the regular quantum vertical trace. BPW's canonical vertical-to-horizontal functor and the BHPW strict tangle/foam functor give

$$(13) \quad \text{Sh}_q : q \text{Tr}(\mathcal{C}; M_R) \longrightarrow \text{Hom}_{R_q}(E_{86}, E_{88}),$$

after the flat-base qHH₀/Chern identification [4, 3]. Here

$$E_{86} = (V^{\otimes 86})_{\lambda=86}, \quad E_{88} = (V^{\otimes 88})_{\lambda=86};$$

the first has rank one and the second rank 88. The tangle maps are $U_q(\mathfrak{sl}_2)$ intertwiners before restriction to these weight spaces.

Base change $q \mapsto 1 + h$ factors through rational localization and the completion of a Noetherian local ring, hence is flat. Reduction modulo h of the trace cokernel simply replaces $q^{|f|}$ by one, giving the ordinary MWW coefficient HH_0 ; right exactness of tensor product suffices for this last assertion.

7.3. Selected class and divided detector. Let $U : P_{86} \rightarrow P_{88}$ be the selected one-cup tangle in the P0 endpoint order and put $T_1 = F^{-1}U$. Define

$$v_T = H_{T_1, T_1}^{-1} (\text{Id}_U \otimes X^{\otimes 227}).$$

The circle counits give $\text{Sh}_h([v_T]) = u_h$. Its absolute degree is

$$-44 + 227 + 315 - 4 = 494.$$

The normalized checked R-matrix on adjacent one-defect basis vectors is

$$\begin{pmatrix} 1 - q^{-2} & q^{-1} \\ q^{-1} & 0 \end{pmatrix}.$$

After the position basis change and $t = q^{-2}$, it is the public positive Burau block; the negative block is its inverse. The physical cable has mixed orientations. BPW's oriented-cabling formula identifies it with the all-positive model up to a writhe-dependent grading shift and pivotal basis maps. The complete public word has writhe zero, so the global shift vanishes. Any degree-zero permutation or pivotal chart P transports the operator, cup and cap simultaneously:

$$W \mapsto PWP^{-1}, \quad u \mapsto Pu, \quad \ell \mapsto \ell P^{-1}.$$

The matrix coefficient is unchanged; we choose the public endpoint labels after this common transport. BPW's cup/cap formulas then imply that the constant terms, after the same position change, are

$$u_0 = e_0 \pm e_5, \quad \ell_0 = e_{87}^* \pm e_2^*.$$

All omitted factors are invertible q -monomials or positive-order h -corrections.

Over $\mathbb{Q}[[h]]$ define

$$D_h = \ell_h(\rho_h(W) - I) \text{Sh}_h.$$

This is well defined on the quantum coefficient trace because (13) has already imposed quantum cyclicity. The exact Magnus calculation and Darné's Andreadakis theorem give $W \in \Gamma_3(P_{44})$ [6, Theorem 6.2]; equivalently, and also by direct evaluation of all 7,744 entries,

$$\rho_h(W) - I \in h^3 \text{End}(E_{88}).$$

Thus D_h takes values in $h^3 \mathbb{Q}[[h]]$. Division by h^3 is unique, and reduction modulo h gives a lift-independent ordinary functional

$$D_3 : HH_0(\mathcal{C}; M_R) \longrightarrow \mathbb{Q}.$$

Positive-order corrections in the cup, cap, and basis change do not alter the cubic coefficient.

The four possible strict pivotal signs give respectively

$$-53824, \quad -53760, \quad -59008, \quad -59072.$$

All are nonzero; the last is the public normalization. Hence

$$(14) \quad D_3([v_T]) = -59072 \neq 0$$

in that normalization, without an unresolved sign premise.

7.4. The complete two-handle cocone. Only D_3 is asserted to descend. Filter the endpoint Temperley–Lieb category by through degree; composition cannot increase this degree. The selected cross-owner one-cup cell has through degree 86. The actual y-passage counts in owner order are

$$(42, 189, 2, 2, 0) \quad \text{for} \quad (m_2, m_3, r_{xy}, r_{yz}, r_{zx}).$$

The P0 normal fields provide arbitrarily many disjoint parallel levels. Taking parallel copies of the same product rectangles defines the Hattori shadow at every cable state. A beta move only changes their placement, and a psi move adds one further parallel negative/positive pair; hence these statewise shadows use one common geometric construction.

For beta, at every cable state apply the P0 collar motion simultaneously to all physical copies. This is the oriented-cabling action of [4, Theorem E], made strict by BHPW. Average the cubic row over the finite set of sign-preserving placements of the selected copies. If the detector uses $a_i^\pm$ copies among $k_i^\pm$ available copies of owner i , this orbit has

$$|\Omega_s| = \prod_i \binom{k_i^-}{a_i^-} \binom{k_i^+}{a_i^+}.$$

It is nonzero on every state on which the row is supported, so division is valid over $\mathbb{Q}$. Constant permutations reindex this average. After a placement is standardized, the remaining pure braid is $I + O(h)$; simultaneous transport of the operator, cup, and cap and $\rho(W) - I = O(h^3)$ show that it changes only terms of order at least four. Thus the cubic average is beta invariant. This

argument does not assume that anti-parallel braid actions factor through a symmetric group.

For an undotted pair addition on any owner meeting the detector gate, the complete action-closed image has through degree at most 84 and vanishes in the degree-86 associated graded. The zero-gate owner is handled by the local core counts. For dotted and undotted additions the whole-source identities are

$$(\epsilon \otimes \epsilon)\Delta(1) = 0, \quad (\epsilon \otimes \epsilon)\Delta(X) = 1.$$

The dotted raw degree +2 is cancelled by the target cabled shift -2 . For a pair addition, forgetting the distinguished new pair has constant-size fibers between the relevant placement orbits. Hence the factor $|\Omega_s|/|\Omega_{s+e_i}|$ converts the target sum to the source sum. These ratios telescope under successive additions, so the target dotted row pulls back to the source row at every level.

These statements are the abstract algebra kernel checked in `Smooth4PC/TransportedPairing.lean`, `Smooth4PC/CubicJet.lean`, and `Smooth4PC/AugmentationCocone.lean`. They prove on the whole cabled direct sum

$$D_3(\beta_i(b)x - x) = 0, \quad D_3(\psi_i^{[0]}x) = 0, \quad D_3(\psi_i^{[1]}x - x) = 0.$$

Consequently D_3 descends through every relation in MWW’s Definition 3.1, and Theorem 3.2 identifies the quotient with the actual W_2 lasagna module. Equation (14) proves that the image of $[v_T]$ is nonzero.

Finally, the detector motion and cap are supported in B . A cobordism outside B is sent by the symmetric-monoidal endpoint functor to an independent tensor factor. Therefore the divided W_2 row has the monoidality used in Section 8.

The finite pairing list, four sign evaluations, source dependencies, and mutation controls are regenerated in `audit/t73_c_comparison_witness.json`; its digest is

410B1C411F321DD83F8ACD76CF5C259D
F2EE341629B13669C2FE458005E5D922.

This completes the proof of Theorem 3.3.

8. THREE-HANDLE SPHERE CLOSURE

We prove Theorem 3.4. Remove the final 4-handle and call the remaining handlebody W_3 . Its boundary is S^3 . Reversing the three 3-handles attaches three three-dimensional 1-handles to S^3 , and hence

$$\partial W_2 \cong \#^3(S^1 \times S^2).$$

The actual attaching sphere system has connected complement, since sphere surgery on it gives the connected boundary of W_3 .

Fix the detector ball B from P0. Choose a small ball in the connected complement of the actual attaching system. Since any two orientation-preserving embedded 3-balls in a connected oriented 3-manifold are ambient isotopic, the actual system may first be isotoped off the fixed ball B . Choose the

standard three nonseparating spheres in the complement of B . Their complement is connected and their classes form the standard basis of $H_2^{\text{sph}}(\partial W_2)$. Horvat–Jabłonowski’s Theorem 5.3 identifies this system with the actual attaching system up to permutation, 3–3 handle slides and isotopy [10]. Applying their relative uniqueness lemma to $Q = \partial W_2 \setminus \text{Int } B$ makes the isotopies fixed on ∂B . The additional boundary slides become tubings to the inessential sphere ∂B after the ball is restored and are removed by a local isotopy. Thus the 3-handles may be represented by standard spheres without moving the detector.

MWW’s intrinsic reformulation of a 3-handle quotient uses the local module action of $\mathcal{S}_0^2(S^2 \times D^2)$. At $N = 2$, let A_0 denote the essential sphere with one dot and A_1 the undotted sphere. The quotient relations are

$$A_0 = 1, \quad A_1 = 0.$$

The symmetric-monoidal naturality required in C applies because the standard spheres lie outside B . For every element v in the complete source,

$$\ell(vA_0) = \ell(v) \text{ev}(\dot{S}^2) = \ell(v), \quad \ell(vA_1) = \ell(v) \text{ev}(S^2) = 0.$$

These are whole-source equations. Hence the detector annihilates the full relation submodule for each of the three standard spheres and descends through the iterated MWW coequalizer. This proves Theorem 3.4 from P0 and C.

The owner lifts, the 32-step Nielsen ledger and the split-tree Lean lemmas are compatible optional coordinate models, but they are not needed for this relative proof. Theorems 3.2 and 3.3 instantiate the required geometry and monoidal detector, so S is unconditional.

9. FINAL IDENTIFICATIONS

MWW Theorem 3.10 identifies the iterated quotient with the module after the three-handles, and Proposition 3.4 says the final four-handle induces a bidegree-(0,0) isomorphism [14]. By Theorem 3.2, this is the handle decomposition of Σ_A^0 . By Theorems 3.3 and 3.4, the specified selected class remains nonzero in

$$\mathcal{S}_{0,(0,494),0}^2(\Sigma_A^0; \mathbb{Q}).$$

MWW Corollary 3.5 gives $\mathcal{S}_0^2(S^4) \cong \mathbb{Z}$, concentrated in bidegree (0,0), so the corresponding rational degree-494 summand vanishes. The intrinsic definition transports lasagna fillings under diffeomorphisms and gives grading-preserving equivalences. Finally, Iwaki’s Proposition 2.1 and (2) prove that the genuine Cappell–Shaneson manifold Σ_A^0 is a homotopy sphere [11]. Applying these general results to the identified handle decomposition proves Theorem 3.5. Theorem 3.3 supplies the nonzero degree-494 class to which these identifications are applied.

10. FORMALIZATION BOUNDARY

The Lean 4 development [7] is valuable precisely because it exposes the logical interface. The theorem in `Smooth4PC/T73Conditional.lean` has the form

$$\text{ExternalGeometry} \longrightarrow \text{CSExternalGeometry} \longrightarrow (\text{HS}(X) \wedge \neg \text{Diff}(X, S^4)).$$

It does not construct either input structure.

The finite files check:

- the matrix A , $A - I$, and their determinants;
- explicit integral inverse formulas for relevant lattice maps;
- the determinant of the proposed sphere coordinate matrix;
- the arithmetic value and nonvanishing of the stored cubic constant;
- the quantum degree 494 from its four recorded integer contributions;
- general linear-algebra quotient and descent lemmas.

The geometric interfaces contain, among others, the predicates

$$\begin{array}{ll} \text{IsActualHHO}, & \text{IsActualSphereMap}, \\ \text{IsActualMWWCoequalizer}, & \text{IsActualFourHandle}. \end{array}$$

These are predicate fields of the ambient universe rather than definitions of smooth topology. Sections 6–9 supply their mathematical instances by public constructions and published theorems. They are not definitions inside Lean, so it would be incorrect to call the Lean development itself a formalization of four-dimensional differential topology or an unconditional kernel proof.

The audit reports that the printed theorem dependencies use only standard logical quotient principles and classical choice, with no `sorryAx`. This is a statement about kernel checking of the algebraic core. P0, C, S and P3 are established mathematically in Sections 6–9, not by the Lean kernel.

11. STATUS TABLE AND REPRODUCIBILITY LEDGER

The table records the current allocation of every external layer. DISCHARGED means that a public generator or primary theorem and an independently checkable argument are both supplied.

Item	Verdict	Reason
P0	DISCHARGED	The public AR source, parameterized product-annulus witness, two geometric cancellations and embedded detector ball identify the replacement presentation with Σ_A^0 .
P1/C	DISCHARGED	The 44 product rectangles give the two-sided Hattori equivalence; BPW/BHPW give the completed endpoint map, and the divided beta/psi cocone has nonzero selected value.
P2/E7	DISCHARGED	P0 supplies the handle pattern and detector ball; HJ and the relative complete-system lemma give standard attaching spheres disjoint from the detector.
P2/E10/S	DISCHARGED	MWW's intrinsic local module action and the monoidal detector give the complete sphere relations for the relative standard system.
P3/E11	DISCHARGED	P0 fixes the actual handle decomposition and MWW Proposition 3.4 gives the bidegree-(0,0) four-handle isomorphism.
P3/E12	DISCHARGED	MWW Corollary 3.5 gives the standard S^4 module concentrated in bidegree zero.
P3/E13	DISCHARGED	Theorem 3.2 identifies Σ_A^0 and Iwaki's determinant criterion applies.

The historical availability script reports the large files as absent. The public replacement witnesses and proofs above discharge every load-bearing role without those files.

12. RELATED WORK

The Cappell–Shaneson literature contains both the original construction [5, 1] and substantial standardness results [2, 9, 12, 11]. Our use of Iwaki is limited to the standard matrix form, the representative table, and the homotopy-sphere criterion. His table entry for (41, 189, 73) is motivation, not a singularity or exoticness theorem.

Manolescu–Neithalath and Manolescu–Walker–Wedrich provide the relevant handle calculus for skein lasagna modules [13, 14]. BPW quantum trace functoriality and the strict BHPW tangle theory provide general comparison tools [4, 3]; Section 7 constructs the candidate-specific comparison with the public endpoint coordinates.

Ren and Willis use Khovanov-theoretic skein lasagna methods to distinguish other exotic four-manifolds [16]. Their computations are independent of the trace-73 construction and are not used as its comparison theorem.

Horvat–Jabłonowski's recent work gives a useful geometric criterion for recognizing three-handle attaching systems [10]. It reduces the replacement problem to boundary, embeddedness, disjointness and homology-basis checks. Applied relative to the detector ball, its complete-system lemma supplies the

standard attaching system used here; the owner-lift and Nielsen ledgers are optional coordinates.

13. REPRODUCIBILITY AND REVIEW

The proof can be replayed in the following order.

- (1) Regenerate the public AR product witness, point-push, Hattori and owner-lift ledgers.
- (2) Verify the full 88-dimensional order-three matrix statement and its mutation controls.
- (3) Compile the Lean cubic, transported-pairing, Frobenius and quotient lemmas.
- (4) Audit Theorem C and the relative-S consequence against the cited BPW/BHPW/MWW functoriality and handle formulas.
- (5) Compile the paper and compare the field-by-field allocation with `T73External.lean`.

The historical large artifacts are unavailable for this replay. Independent review should check the public AR mapping-torus diffeomorphism, product framings and two cancellations, the BPW coefficient comparison in Theorem 3.3, and its monoidality used by the relative sphere theorem.

14. CONCLUSION

The public AR witness identifies the manifold, the product Hattori comparison turns the finite Burau anomaly into a genuine MWW divided functional, and the relative sphere theorem carries its nonzero degree-494 class through the three- and four-handles. The standard S^4 module has no such degree. Therefore Theorem 3.6 gives a counterexample to the smooth four-dimensional Poincare conjecture.

APPENDIX A. RETIRED HISTORICAL OBJECTS

The following table records historical artifacts cited by earlier project notes. They are unavailable and no longer supply any current input: the AR, coefficient-comparison, and relative-sphere witnesses discharge the old P0/P1/P2 roles. The digests are retained for provenance.

Priority	Object	SHA-256
P0	cs_presentation.py	B20BDF6FBF2A0E629083E2E62DD1DDF9172A4E8893311CE122FFC4FEB388DFA6DF4D7906ADB18BA2D3237D4B49F8985A5B18D8C39FF522AD6498F2369803FCBE6912A64457557469E5C691B4D57ABDBBF4C45ADB05492777C574223D0C06F8A4E1C4564ED98F30EAEBOA1AEAAD95ACA
P0	t73_eps0.erkmo.json	B5AD7B226C79B7D5BF3F259E8B56F5412D9E4760F836D9E04E6E0024CAE03A3DBB49A74FF5D129756F25450B30F10F7CB2E97EB0981036768B8F1C1D5EF16F8817EC1A7B951B6B32DBA2F9211FFD725A13AED414447ED33057C99D340910B50109983CF36874C14ECE0E6A2834A12C249D93A6CB320200A906F461EF3131127EA40C80A9A7828128E6F28048084170567F3D3618D6A790A9B60EE8085B647AC2AB742E1BC9C15841F1BEF015034217B5DE1AC479699EC79DE76D4265993DE437493A3AAA6CABB636F98998644BF3181CEE620E6B085A5F9E1C73CFDD1AD04FC0682CEC74DA3DBF8AFE70DD19C038E3A04D1B627C0343A1C464319704EAFADCA127902A2E4E90CF1C283004359B7ADC24EABF67C0D1AE309F0281297710A283B8ED5A11D88C8E810837932313F4C53227
P0	t73_reduced_billiard.pd.json	
P0	CUT_OBJECT.json	
P0	BASE_INTERACTION_SUPPORT.json	
P0	NORMAL_FIELD_MOVIE_CERT.json	
P0	T73_COLLAR_BRAID.json	
P0	verify_t73_collar_braid.py	
P1	ACTUAL_PD_CABLE_UNIT_CERT.json	
P1	PRODUCT_NORMAL_CHRISTOFFEL_THXY_MOVIE.json	
P2	TH1_EXHAUSTIVE_ALL_ROW_GEOMETRY.json	
P2	TH2_EXHAUSTIVE_ALL_ROW_GEOMETRY.json	
P2	THXY_FULL_MACRO_P3FREE_HJ_CERT.json	

The same values and candidate paths are machine-readable in `audit/geometric_evidence_manifest.json`. These objects are not included. The public AR replacement discharges P0 without them. A proof of C must still recover or replace the actual coefficient/Hattori object and its functorial role. The historical P2 objects are no longer required by the relative sphere argument.

APPENDIX B. CORRESPONDENCE WITH THE LEAN FIELDS

For readers checking the formal boundary, the principal allocation is:

Lean field	Mathematical obligation
<code>x0, ell0, ell0_x0</code>	P1 comparison and the selected genuine MWW class.
<code>q01, ell1_comp_q01</code>	Complete two-handle beta/psi quotient and descent.
<code>q12, ell2_comp_q12</code>	Actual E7 spheres, MWW hemisphere maps, and complete coequalizer.
<code>transport, fourIso</code>	E11 applied to the same graded lasagna module.
<code>s4DegreeZero</code>	E12 after identifying G with rational $N = 2$ lasagna.
<code>diffeomorphismEquiv</code>	Graded diffeomorphism invariance of that same object.
<code>matrixConditionsToHomotopySphere</code>	E13 for the manifold identified by P0.

These are exactly the candidate-level obligations isolated in Sections 5–8. They have not been discharged. The Lean theorem consumes no stronger statement, but it still requires them as structure fields.

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